

Homework 8

Please read Section 1.6 of Durrett through the end of Example 1.27.

Please also read Section 2.2 of Durrett (binomial and multinomial distributions) through the end of Example 2.16. Theorem 2.2 tells you the expected value (mean) and variance of a binomially distributed random variable. For now, you can just read the statement of the theorem. We will come back to this calculation in Chapter 6, when we will learn how to get these values quickly and without the heavy algebra that Durrett has to use in Chapter 2.

Please work the following exercises **from Chapter 1**:

43 (answer from back of book: $-17/216$; please show how to calculate this number)

We play a game where we roll three dice and win \$1 for each 6 we roll. If we roll no 6's, we lose \$1. The random variable X representing our payout takes values $-1, 1, 2, 3$. We calculated the probabilities of rolling 0, 1, 2, 3 6's in Quiz 3. Those calculations give us the following:

no 6's: $P(X = -1) = 125/216$

one 6: $P(X = 1) = 75/216$

two 6's: $P(X = 2) = 15/216$

three 6's: $P(X = 3) = 1/216$

We thus have $E[X] = (-1)125/216 + (1)75/216 + (2)15/216 + (3)1/216 = (-125+75+45+3)/216 = -17/216$. We expect to lose \$17/216 on average in a round of playing this game.

We could also calculate the probabilities above using the general rule for binomial probabilities. The random variable counting the number of 6's rolled has the binomial(3, 1/6) distribution.

50 (Durrett uses "mean" as a synonym for "expected value")

The values of the random variable X are 1, 2, 3, 4, 5. Durrett tells us the distribution of X :

$P(X = 1) = 1/15$

$P(X = 2) = 2/15$

$P(X = 3) = 3/15$

$P(X = 4) = 4/15$

$P(X = 5) = 5/15$

We thus have $E[X] = 1(1/15) + 2(2/15) + 3(3/15) + 4(4/15) + 5(5/15) = 55/15 = 11/3$

To calculate the variance, we use the shortcut formula $\text{Var}(X) = E[X^2] - E[X]^2$. By the Law of the Unconscious Statistician (LOTUS), we have

$$E[X^2] = 1^2(1/15) + 2^2(2/15) + 3^2(3/15) + 4^2(4/15) + 5^2(5/15) = 225/15 = 15.$$

$$\text{Therefore, } \text{Var}(X) = 15 - (11/3)^2 = 15 - 121/9 = (135 - 121)/9 = 14/9 \approx 1.56.$$

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The possible values for X are 28, 30, and 31.

$$P(X = 28) = 1/12 \text{ (just February)}$$

$$P(X = 30) = 4/12 \text{ (April, June, September, November)}$$

$$P(X = 31) = 7/12 \text{ (January, March, May, July, August, October, December)}$$

$$E[X] = 28(1/12) + 30(4/12) + 31(7/12) = 365/12 \approx 30.4 \text{ days. (Oh! That makes sense. The average length of a month is the number of days in the year divided by the number of months.)}$$

$$\text{By LOTUS, } E[X^2] = 28^2(1/12) + 30^2(4/12) + 31^2(7/12) = 11111/12$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = 11111/12 - (365/12)^2 = 107/144 \approx 0.74. \text{ The standard deviation } \sigma(X) \text{ is about 0.86 days.}$$

55 (answer from back of book: “no.” Explain why there cannot be such a random variable.)

Suppose we had a random variable X with $E[X] = 3$ and $E[X^2] = 8$. We would then have $\text{Var}(X) = 8 - 3^2 = -1$. But that is impossible, since the variance of X would also be $E[(X - 3)^2]$, which must be non-negative, since the random variable $(X - 3)^2$ is non-negative. (Squares of real numbers are non-negative.)

Please work the following exercises **from Chapter 2**:

28 (Let N be the number of 3's in 8 rolls. Match this situation with the binomial-distribution story to get $N \sim \text{binomial}(n, p)$ for suitable n and p. Use the formula for binomial probabilities to calculate $P(N = 2)$.)

A die is rolled 8 times. Let N be the random variable counting the number of 3's in the 8 rolls. Assuming a fair die, the probability of a 3 on any roll is 1/6. Therefore, assuming the rolls are independent, we have $N \sim \text{binomial}(8, 1/6)$. According to the general formula for binomial probabilities, we have

$$P(N=2) = C_{8,2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^6 = 28 \cdot \frac{15625}{1679616} \approx 0.26.$$

29 (answer from the back of the book: 90/1024. To get Durrett's answer, assume that Mary is guessing on the last five questions. Let N be the number of questions that Mary answers

correctly. Match the story in this question with the binomial distribution story to get $N \sim \text{binomial}(n, p)$ for suitable n and p . Use the formula for binomial probabilities to calculate $P(N = 3)$.)

Mary has a probability of $1/4$ of succeeding at guessing each of the last five questions correctly. Since she is guessing, it is reasonable to assume that the outcomes for these questions are independent. Therefore the random variable N counting the number of correct guesses has the $\text{binomial}(5, 1/4)$ distribution. According to the general formula for binomial probabilities, we have

$$P(N=3) = C_{5,3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^2 = 10 \cdot \frac{9}{1024} \approx 0.0879.$$

31 part (a) only (answer from the back of the book: 0.152. Let N be the number of unmarried women in the random sample. Match this situation with the binomial-distribution story to get $N \sim \text{binomial}(n, p)$ for suitable values of n and p . Use the formula for binomial probabilities to calculate $P(N = 10)$.)

Each randomly selected person has a 32.6% chance of being unmarried. We will treat the selections as independent. There are two ways to justify that independence assumption:

- The selections are independent if the samples are made with replacement so that a woman in the population might be picked more than once.
- If the selections are made without replacement—the 30 people are a permutation of 30 drawn from the set of people who gave birth—then the selections are not independent. However, since the population is so large compared to the sample size, they are practically independent.

Assuming independence, the number N of unmarried women in the sample is a random variable with the $\text{binomial}(30, 0.326)$ distribution. Therefore,

$$P(N=10) = C_{30,10} (0.326)^{10} (1-0.326)^{20} \approx 0.152.$$